

GATE (DA) Data Science and Artificial Intelligence

Full-Length Mock Examination

Duration: 180 Minutes | Total Marks: 100 | Total Questions: 65

Instructions

- This examination paper contains two sections: General Aptitude (GA) (Questions 1–10) and Technical Section (Questions 11–65).
 - Marking Distribution: Questions 1 to 5 carry 1 mark each; Questions 6 to 10 carry 2 marks each (Total GA = 15 Marks). Questions 11 to 35 carry 1 mark each; Questions 36 to 65 carry 2 marks each (Total Technical = 85 Marks).
 - Question Types: MCQ (Multiple Choice Questions): only one choice is correct. Negative marking applies: 1-mark MCQ: deduct $\frac{1}{3}$ mark for an incorrect response; 2-mark MCQ: deduct $\frac{2}{3}$ mark for an incorrect response.
 - MSQ (Multiple Select Questions): one or more choices are correct. No negative marking; no partial marking.
 - NAT (Numerical Answer Type): numerical value must be entered. No negative marking.
 - No credit is awarded or deducted for unattempted questions.
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Section I: General Aptitude (GA)

Q.1 - Q.5 carry 1 mark each

Q.1 [MCQ | 1 Mark]

Choose the word that is opposite in meaning to the word underlined:

The diplomat was widely lauded for her equanimity during the geopolitical crisis.

- (A) Composure
- (B) Agitation
- (C) Reticence
- (D) Indifference

Q.2 [MCQ | 1 Mark]

Which of the following sentences is grammatically correct?

- (A) Neither the principal nor the teachers was present at the annual convocation.
- (B) Neither the principal nor the teachers were present at the annual convocation.
- (C) Neither the teachers nor the principal were present at the annual convocation.
- (D) Neither the principal or the teachers was present at the annual convocation.

Q.3 [MCQ | 1 Mark]

If $x = \sqrt{7 + 4\sqrt{3}}$, then the value of $x + 1/x$ is:

- (A) 4
- (B) $2\sqrt{3}$
- (C) $4\sqrt{3}$
- (D) 2

Q.4 [MCQ | 1 Mark]

A regular polygon has an interior angle that is 140° . How many sides does this polygon have?

- (A) 7
- (B) 8
- (C) 9
- (D) 10

Q.5 [MCQ | 1 Mark]

Identify the figure that replaces the question mark (?) in the sequence:

A cube is folded from a flat cross-shaped paper template. Opposite to the face marked with the symbol $\bullet$ is a face marked with:

- (A) $\blacksquare$
- (B) $\star$
- (C) $\triangle$
- (D) $\odot$

(Assume the net has consecutive horizontal faces: $\blacksquare$, $\odot$, $\star$, $\triangle$ with $\bullet$ hinged on top of $\odot$ and $\oplus$ hinged below $\odot$)

Q.6 - Q.10 carry 2 marks each**Q.6 [MCQ | 2 Marks]**

Four friends—A, B, C, and D—sit in a row. Exactly one of them is lying.

- A says: "B is to my immediate right."
- B says: "C is not sitting at either end."
- C says: "D is sitting to my left."
- D says: "A is telling the truth."

Who is the liar?

- (A) A
- (B) B
- (C) C
- (D) D

Q.7 [NAT | 2 Marks]

A train running at 72 km/h crosses a pole in 12 seconds. The time taken by the train to cross a bridge of length 360 m is _____ seconds.

Q.8 [MCQ | 2 Marks]

In a class of 100 students, 60 like Python, 45 like R, and 20 like neither. How many students like both Python and R?

- (A) 15
- (B) 25
- (C) 30
- (D) 35

Q.9 [MCQ | 2 Marks]

Given that a, b, c are positive real numbers such that $a + b + c = 1$, the maximum value of a^2b^3c is achieved when the ratio $a:b:c$ is:

- (A) 1:1:1
- (B) 2:3:1
- (C) 3:2:1
- (D) 4:9:1

Q.10 [MCQ | 2 Marks]

Consider the statement: "All models are useful, but some models are wrong."

Which of the following is logically equivalent to the negation of this statement?

- (A) At least one model is not useful, or all models are correct.
- (B) No models are useful, and all models are wrong.
- (C) Every model is useless, or no model is wrong.
- (D) At least one model is not useful, and all models are correct.

Section II: Technical Section (Questions 11 to 65)

Q.11 - Q.35 carry 1 mark each

Q.11 [MCQ | 1 Mark]

Let A be a 3×3 matrix with real entries such that $\det(A) = 4$. What is the value of $\det(2A^{-1} \text{adj}(A))$?

- (A) 8
- (B) 32
- (C) 64
- (D) 128

Q.12 [NAT | 1 Mark]

Let X be a continuous random variable uniformly distributed over the interval $[0, 4]$. The conditional probability $P(X > 2.5 \mid X > 1)$ is _____ (round off to 2 decimal places).

Q.13 [MCQ | 1 Mark]

Let $R(A, B, C, D, E)$ be a relation with functional dependencies:

$$F = \{A \rightarrow B, BC \rightarrow D, E \rightarrow C\}$$

Which of the following is a candidate key of R ?

- (A) AE
- (B) ACE
- (C) ABE
- (D) ADE

Q.14 [MSQ | 1 Mark]

Which of the following statements about Support Vector Machines (SVM) is/are TRUE?

- (A) The optimal hyperplane depends only on the support vectors.
- (B) Using the soft-margin parameter $C \rightarrow \infty$ allows zero training errors on linearly separable data.
- (C) SVM with an RBF kernel has an infinite-dimensional feature space.
- (D) SVM decision boundary is sensitive to outliers that are not support vectors.

Q.15 [MCQ | 1 Mark]

What is the time complexity of searching for an element in a balanced Binary Search Tree (AVL tree) containing n elements?

- (A) $O(1)$
- (B) $O(\log n)$
- (C) $O(n)$
- (D) $O(n \log n)$

Q.16 [NAT | 1 Mark]

Consider the function $f(x, y) = x^2 + 2y^2 - 4x - 4y + 7$. The minimum value of $f(x, y)$ over $\mathbb{R}^2$ is _____.

Q.17 [MCQ | 1 Mark]

In Linear Discriminant Analysis (LDA) for binary classification, which assumption is explicitly made?

- (A) Classes have identical mean vectors.
- (B) Features are conditionally independent given the class label.
- (C) Both classes follow a Gaussian distribution with identical covariance matrices.
- (D) The decision boundary is always non-linear.

Q.18 [MCQ | 1 Mark]

Consider the recurrence relation:

[Expression/formula not present in the supplied source text]

Using asymptotic notation, $T(n)$ is:

- (A) $\Theta(n \log n)$
- (B) $\Theta(n \log^2 n)$
- (C) $\Theta(n^2)$
- (D) $\Theta(n)$

Q.19 [MSQ | 1 Mark]

Which of the following statements is/are TRUE for lossless-join decomposition?

- (A) A decomposition of relation R into R_1 and R_2 is lossless if and only if $R_1 \cap R_2 \rightarrow R_1$ or $R_1 \cap R_2 \rightarrow R_2$ holds in F^+ .
- (B) Every dependency-preserving decomposition is guaranteed to be lossless.
- (C) BCNF decomposition is always guaranteed to be lossless.
- (D) 3NF synthesis algorithm guarantees both lossless join and dependency preservation.

Q.20 [NAT | 1 Mark]

Two unbiased fair 6-sided dice are rolled simultaneously. The probability that the sum of the outcomes is an odd prime number is _____ (round off to 3 decimal places).

Q.21 [MCQ | 1 Mark]

Consider the following relational schema:

Employee(emp_id, emp_name, dept_id, salary)

What does the following query return?

```
SELECT dept_id, COUNT(*)  
FROM Employee  
WHERE salary > 50000  
GROUP BY dept_id  
HAVING COUNT(*) > 2;
```

- (A) All departments having more than 2 employees.
- (B) All departments having more than 2 employees where each of those employees earns more than 50000.
- (C) All departments where total employee salary exceeds 100000.
- (D) The count of all employees earning over 50000 across all departments.

Q.22 [MCQ | 1 Mark]

Let A be an $n \times n$ symmetric matrix. Which of the following is NOT necessarily true?

- (A) All eigenvalues of A are real.
- (B) Eigenvectors corresponding to distinct eigenvalues are mutually orthogonal.
- (C) A is invertible.
- (D) A can be orthogonally diagonalized.

Q.23 [NAT | 1 Mark]

A classifier predicts labels on a test set of 100 samples. The confusion matrix is:

- True Positive (TP) = 40
- False Positive (FP) = 10
- False Negative (FN) = 20
- True Negative (TN) = 30

The Precision of this classifier is _____ (round off to 2 decimal places).

Q.24 [MCQ | 1 Mark]

Breadth-First Search (BFS) is executed on a directed graph starting from vertex s . The shortest path found to any vertex v represents the shortest path in terms of:

- (A) Total sum of edge weights.
- (B) Minimum number of edges.
- (C) Product of edge capacities.
- (D) Maximum bottleneck capacity.

Q.25 [MSQ | 1 Mark]

Which of the following statements about Logistic Regression is/are correct?

- (A) Logistic regression uses cross-entropy loss because mean squared error is non-convex for logistic regression.
- (B) The decision boundary of standard logistic regression is linear in the feature space.
- (C) L1 regularization (Lasso) in logistic regression can drive some weight coefficients to exactly zero.
- (D) Logistic regression assumes that features are statistically independent.

Q.26 [NAT | 1 Mark]

Let $X \sim \text{Poisson}(\lambda)$. If $P(X = 1) = 2 P(X = 2)$, then the variance of X is _____.

Q.27 [MCQ | 1 Mark]

Consider an array $A = [2, 5, 8, 12, 16, 23, 38, 56, 72, 91]$. If classical Binary Search is executed to find key $k = 23$, how many comparisons are made?

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Q.28 [MCQ | 1 Mark]

Let $f(x) = |x|^3$ for $x \in \mathbb{R}$. At $x = 0$, the function $f(x)$ is:

- (A) Continuous but not differentiable.
- (B) Differentiable once, but its second derivative does not exist.
- (C) Twice differentiable, but its third derivative does not exist.
- (D) Infinitely differentiable.

Q.29 [NAT | 1 Mark]

Let $S = \{1, 2, 3, \dots, 100\}$. A number is drawn at random from S . The probability that the number is divisible by 4 or 6 is _____ (round off to 2 decimal places).

Q.30 [MCQ | 1 Mark]

In Decision Tree learning, Gini Impurity for a binary classification node with class proportions p and $1 - p$ is given by:

- (A) $-p \log_2 p - (1-p) \log_2 (1-p)$
- (B) $2p(1 - p)$
- (C) $1 - \max(p, 1-p)$
- (D) $p^2 + (1-p)^2$

Q.31 [MSQ | 1 Mark]

Which of the following algorithms have a worst-case time complexity of $O(n \log n)$ for sorting n real numbers?

- (A) QuickSort
- (B) MergeSort
- (C) HeapSort
- (D) BubbleSort

Q.32 [NAT | 1 Mark]

Let matrix $M = \begin{bmatrix} 1 & 4 \\ 2 & 3 \end{bmatrix}$ (rows: 1 4 / 2 3). The absolute difference between the two eigenvalues of M is _____.

Q.33 [MCQ | 1 Mark]

Under the Naive Bayes assumption, features $X_1, X_2, \dots, X_d$ are assumed to be:

- (A) Mutually independent across all classes unconditionally.
- (B) Pairwise independent given the class label Y .
- (C) Linearly dependent on the class label Y .
- (D) Normally distributed with zero covariance under every circumstance.

Q.34 [MCQ | 1 Mark]

In an SQL database, what is the evaluated result of the expression:

$(NULL = NULL)$ and $(NULL IS NULL)$?

- (A) TRUE and TRUE
- (B) UNKNOWN and TRUE
- (C) FALSE and TRUE
- (D) UNKNOWN and UNKNOWN

Q.35 [NAT | 1 Mark]

Let $Z \sim N(0, 1)$ be standard normal. If $\Phi(1.645) = 0.95$, then $P(-1.645 \leq Z \leq 1.645)$ is _____ (round off to 2 decimal places).

Q.36 - Q.65 carry 2 marks each**Q.36 [MCQ | 2 Marks]**

Let X and Y have joint probability density function:

[Expression/formula not present in the supplied source text]

The value of constant c is:

- (A) $1/2$
- (B) $2/3$
- (C) 1
- (D) $3/2$

Q.37 [NAT | 2 Marks]

For the random variables X and Y in Q.36 with the calculated constant c , the marginal expectation $E[X]$ is _____ (round off to 3 decimal places).

Q.38 [MSQ | 2 Marks]

Let $A \in \mathbb{R}^{m \times n}$ with rank r . Let nullity of A be denoted by $\text{null}(A)$. Which of the following statements must be TRUE?

- (A) $r + \text{null}(A) = n$
- (B) The column space of A and the null space of A^T are orthogonal complements in $\mathbb{R}^m$.
- (C) If $m = n$ and $r = n$, then $Ax = b$ has a unique solution for every $b \in \mathbb{R}^n$.
- (D) $\text{rank}(A^T A) = \text{rank}(A A^T) = r$

Q.39 [NAT | 2 Marks]

A binary classification test for a rare disease has a sensitivity (true positive rate) of 95% and a specificity (true negative rate) of 90%. If the disease prevalence in the population is 2%, what is the posterior probability (in percentage) that a randomly chosen person who tests positive actually has the disease? _____ (round off to 2 decimal places).

Q.40 [MCQ | 2 Marks]

Consider relation schema $R(U, V, W, X, Y, Z)$ with functional dependencies:

[Expression/formula not present in the supplied source text]

What is the highest normal form satisfied by relation R ?

- (A) 1NF
- (B) 2NF
- (C) 3NF
- (D) BCNF

Q.41 [MCQ | 2 Marks]

Consider the following pseudocode:

```
def mystery(n):
    c = 0
    i = 1
    while i < n:
        j = 1
        while j < n:
            c += 1
            j = j * 2
        i = i * 3
    return c
```

What is the worst-case asymptotic time complexity of $\text{mystery}(n)$?

- (A) $\Theta(\log n)$
- (B) $\Theta((\log n)^2)$
- (C) $\Theta(n \log n)$
- (D) $\Theta(n^2)$

Q.42 [NAT | 2 Marks]

A random variable X has variance $\sigma^2 = 25$. Using Chebyshev's inequality, the upper bound on $P(|X - \mu| \geq 15)$ is _____ (round off to 3 decimal places).

Q.43 [MSQ | 2 Marks]

Which of the following statements regarding the Bias-Variance trade-off is/are TRUE?

- (A) Increasing the value of k in k -Nearest Neighbors increases variance and decreases bias.
- (B) Pruning a fully-grown Decision Tree decreases variance while increasing bias.
- (C) Increasing the regularization hyperparameter λ in Ridge Regression increases bias.
- (D) A high training error combined with a high test error typically indicates high bias.

Q.44 [MCQ | 2 Marks]

Let S be a set of 5 distinct elements. How many reflexive and symmetric binary relations can be defined on S ?

- (A) 2^{10}
- (B) 2^{15}
- (C) 2^{20}
- (D) 2^{25}

Q.45 [NAT | 2 Marks]

Consider the matrix:

[Expression/formula not present in the supplied source text]

The trace of A^{-1} is _____ (round off to 2 decimal places).

Q.46 [MCQ | 2 Marks]

Consider relations $R(A, B)$ and $S(B, C)$. R contains 200 tuples, and S contains 300 tuples. The attribute B is a primary key of S and a foreign key in R referencing $S(B)$. What are the minimum and maximum possible number of tuples in the natural join $R \bowtie S$?

- (A) Min = 0, Max = 200
- (B) Min = 200, Max = 200
- (C) Min = 0, Max = 300
- (D) Min = 200, Max = 60000

Q.47 [NAT | 2 Marks]

In a dataset with two classes (Positive and Negative), a decision tree split creates two child nodes:

- Node 1: 30 Positive, 10 Negative
- Node 2: 10 Positive, 50 Negative

The weighted average Gini Impurity of this split is _____ (round off to 3 decimal places).

Q.48 [MCQ | 2 Marks]

Let $f(x) = x^3 - 3x + 2$. Over the closed interval $[-2, 2]$, the global maximum value of $f(x)$ is:

- (A) 0
- (B) 2
- (C) 4
- (D) 8

Q.49 [MSQ | 2 Marks]

Let $G = (V, E)$ be an undirected, connected, weighted graph with distinct positive edge weights. Which of the following statements is/are TRUE?

- (A) The Minimum Spanning Tree (MST) of G is unique.
- (B) The second-best minimum spanning tree differs from the MST by at least one edge.
- (C) The shortest path between any two vertices u and v is always part of the MST.
- (D) If the weight of the lightest edge in G is reduced, the MST does not change.

Q.50 [NAT | 2 Marks]

Let $X_1, X_2, \dots, X_{100}$ be independent and identically distributed random variables from a distribution with mean $\mu = 10$ and variance $\sigma^2 = 16$. Let $\bar{X} = (1/100) \sum X_i$ ($i = 1$ to 100).

Using the Central Limit Theorem, the value of $P(\bar{X} \geq 10.4)$ is approximately _____ (round off to 3 decimal places).

(Given: $\Phi(1.0) = 0.8413$)

Q.51 [MCQ | 2 Marks]

Consider the relations:

Students(sid, sname, gpa)

Enrollment(sid, cid)

Which of the following SQL queries selects the names of students who have enrolled in every course offered in the Courses(cid) table?

(A)

```
SELECT sname FROM Students S
WHERE NOT EXISTS (
    SELECT cid FROM Courses
    EXCEPT
    SELECT cid FROM Enrollment E WHERE E.sid = S.sid
);
```

(B)

```
SELECT sname FROM Students S
WHERE EXISTS (
    SELECT cid FROM Courses
    INTERSECT
    SELECT cid FROM Enrollment E WHERE E.sid = S.sid
);
```

(C)

```
SELECT sname FROM Students S JOIN Enrollment E ON S.sid = E.sid
GROUP BY S.sid, S.sname
HAVING COUNT(E.cid) = (SELECT COUNT(*) FROM Students);
```

(D)

```
SELECT sname FROM Students S
WHERE cid IN (SELECT cid FROM Courses);
```

Q.52 [NAT | 2 Marks]

Let $A = \begin{bmatrix} 3 & 1 \\ 0 & 3 \end{bmatrix}$ (rows: 3 1 / 0 3). The $(1, 2)$ -th entry of the matrix A^{10} is _____.

Q.53 [MCQ | 2 Marks]

We wish to test $H_0: \mu = 50$ against $H_1: \mu > 50$ for a normal population with known variance $\sigma^2 = 25$ using a sample of size $n = 25$. If the rejection region is $\bar{X} \geq 51.645$, what is the probability of a Type II error (β) when the true population mean is $\mu = 53$?

(Given: $\Phi(1.355) = 0.9123$, $\Phi(-1.355) = 0.0877$)

- (A) 0.0877
- (B) 0.0500
- (C) 0.9123
- (D) 0.0250

Q.54 [MSQ | 2 Marks]

Consider the k-Nearest Neighbors (k-NN) algorithm. Which of the following statements is/are correct?

- (A) As $k \rightarrow n$ (where n is total training points), the classifier tends to predict the majority class in the dataset.
- (B) For $k = 1$, the training error on the training dataset itself is always 0 (assuming no conflicting identical points).
- (C) Euclidean distance is robust and remains effective without feature scaling even if one feature dominates in magnitude.
- (D) k-NN is a parametric model because k is a model parameter.

Q.55 [NAT | 2 Marks]

Let $f(x) = e^x - 2x$. Using gradient descent with learning rate $\eta = 0.1$, starting at $x_0 = 0$, what is the updated value x_1 ? _____.

Q.56 [MCQ | 2 Marks]

Consider a hash table of size $m = 11$ using open addressing with double hashing:

$$h(k, i) = (h_1(k) + i \cdot h_2(k)) \bmod 11$$

where $h_1(k) = k \bmod 11$ and $h_2(k) = 1 + (k \bmod 7)$.

After inserting keys 18, 29, 40, at which index will key 51 be inserted?

- (A) 7
- (B) 8
- (C) 9
- (D) 10

Q.57 [NAT | 2 Marks]

The directional derivative of the scalar field $\phi(x, y, z) = x^2y + y^2z + z^2x$ at the point $P(1, 1, 1)$ in the direction of the unit vector $u = (1/\sqrt{3})(1, 1, 1)^T$ is _____ (round off to 2 decimal places).

Q.58 [MSQ | 2 Marks]

Let X and Y be two random variables with finite second moments. Which of the following statements is/are ALWAYS true?

- (A) If $\text{Cov}(X, Y) = 0$, then X and Y are independent.
- (B) If X and Y are independent, then $\text{Var}(X - Y) = \text{Var}(X) + \text{Var}(Y)$.
- (C) $-1 \leq \text{Corr}(X, Y) \leq 1$.
- (D) $\text{Var}(aX + b) = a^2 \text{Var}(X)$ for any real constants a, b .

Q.59 [NAT | 2 Marks]

Consider the following recursive algorithm:

```
def solve(n):
    if n <= 1:
        return 1
    return solve(n - 1) + solve(n - 2)
```

How many total function calls to solve are executed to compute $\text{solve}(5)$? _____.

Q.60 [MCQ | 2 Marks]

In a binary logistic regression model, the learned weight vector is $w = [0.5, -1.0]^T$ and bias $b = 0$. The sigmoid activation is $\sigma(z) = 1 / (1 + e^{-z})$. For an input instance $x = [2.0, 1.0]^T$, what is the predicted probability $P(Y = 1 | x)$?

- (A) 0.25
- (B) 0.50
- (C) 0.73
- (D) 0.88

Q.61 [NAT | 2 Marks]

Consider relation $R(A, B, C, D, E, F)$ with functional dependencies:

[Expression/formula not present in the supplied source text]

The number of attributes in the closure $(AD)^+$ is _____.

Q.62 [MCQ | 2 Marks]

Consider the following dynamic programming recurrence:

[Expression/formula not present in the supplied source text]

This recurrence solves which classic computational problem?

- (A) Fractional Knapsack
- (B) 0/1 Knapsack
- (C) Longest Common Subsequence
- (D) Matrix Chain Multiplication

Q.63 [NAT | 2 Marks]

Let the matrix $A = \begin{bmatrix} 1 & 2 \\ 2 & 4 \end{bmatrix}$ (rows: 1 2 / 2 4). The largest singular value (σ_1) of A is _____ (round off to 1 decimal place).

Q.64 [MSQ | 2 Marks]

Which of the following statements about Depth-First Search (DFS) on a directed graph is/are TRUE?

- (A) A directed graph contains a cycle if and only if DFS yields at least one back edge.
- (B) In a Directed Acyclic Graph (DAG), vertices ordered by decreasing finish times yield a valid topological sort.
- (C) Cross edges can connect vertices across two disjoint DFS trees.
- (D) The discovery time of vertex u is always strictly smaller than that of vertex v if there is an edge (u, v) .

End of Examination Paper

Q.65 [NAT | 2 Marks]

Evaluate the definite integral:

[Expression/formula not present in the supplied source text]

The value of I is _____ (round off to 3 decimal places).